Decentralised Auction-Based Peer-to-Peer Energy Management for a Hybrid Solar-Grid Agricultural Dryer Microgrid Using ESP32 Embedded Controllers

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Summary. Farmers who share a dryer facility rarely have enough electricity to run every dryer at once, so power goes to whoever arrives first rather than to the produce most at risk of spoiling. This project designs, simulates and builds a three-layer microgrid. In it, two dryer controllers and one power controller (all ESP32 boards) agree every 10 s on which dryer receives power, using each chamber's measured humidity as its bid. The boards talk directly over ESP-NOW, so no router, internet or cloud server is needed. A DPDT relay switches the supply between grid and solar automatically and falls back to grid if the controller fails. The full-scale design (1.41 kW PV, 220 Ah battery) was evaluated in MATLAB/Simulink, and the control system was built and tested as a bench prototype. The dryer is the first application, not the limit: the same mechanism can share scarce power among any set of loads (a hospital, school, homes and places of worship in one community) by priority, and keep them supplied locally when the national grid fails.
Key results: Jain fairness index **0.93** (8-hour simulation) and **0.90** (live, on hardware); **100 %** solar/battery supply in simulation; all three boards linked and agreeing on the winner in every observed round; fail-safe grid↔solar switching verified on the bench.

1 Problem Statement

Nigeria loses an estimated US$9 billion of harvested produce every year, much of it through poor drying and storage, and erratic power supply is a major cause [2]. Solar and hybrid solar dryers dry produce faster and more hygienically than open-sun drying [3, 4]. Where several farmers share one dryer facility, however, the available power is usually less than the total demand, and three problems follow:

(i) **No priority by need.** Power is allocated by arrival time or argument, so the wettest produce may wait and spoil.

(ii) **No trustworthy rule.** Users can overstate urgency, which leads to disputes.

(iii) **Existing smart systems need infrastructure.** Peer-to-peer (P2P) energy-sharing schemes rely on cloud platforms, internet-connected meters or blockchain ledgers [5, 6, 7], which are unreliable or unaffordable in rural communities.

The same problem exists at community scale. Nigeria's national grid collapsed 12 times in 2024 alone [10], and whenever supply is short, a whole community (hospital, school, homes, mosques and churches) competes for whatever local power remains, with no agreed rule for who comes first.

Research gap: a low-cost, offline method that divides scarce power between shared loads *fairly* and *according to measured need*, validated on real hardware rather than in software alone. Agricultural dryers are the test case. The work builds on the author's earlier ESP32-based peer-to-peer energy-sharing platform for residential communities [1], extending it from equal sharing among tenants to need-based, fully offline allocation between productive loads.

2 Aim, Objectives and Performance Targets

Aim: design, implement and evaluate a decentralised P2P energy management system for a hybrid solar-grid agricultural dryer, using a dual-path priority mechanism (sensor-verified humidity plus a manual re-check button) to allocate scarce energy fairly and by need.

O1. Design a three-layer architecture (power, control, dryer endpoints) with source-selection logic for solar and grid.

O2. Implement a decentralised ESP-NOW broadcast allocation protocol on ESP32 boards with sensor-driven bids and a manual override.

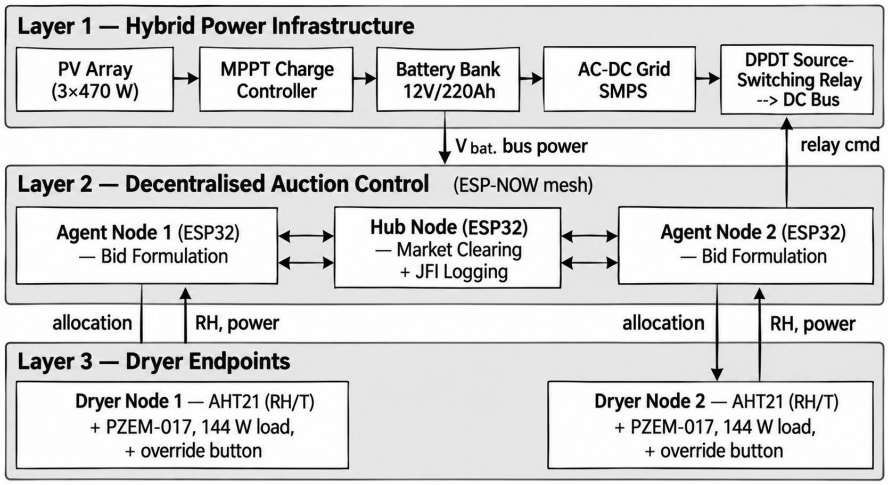
O3. Evaluate performance in MATLAB/Simulink and on a physical prototype against the targets in Table 1.

Table 1: Key performance indicators (KPIs).

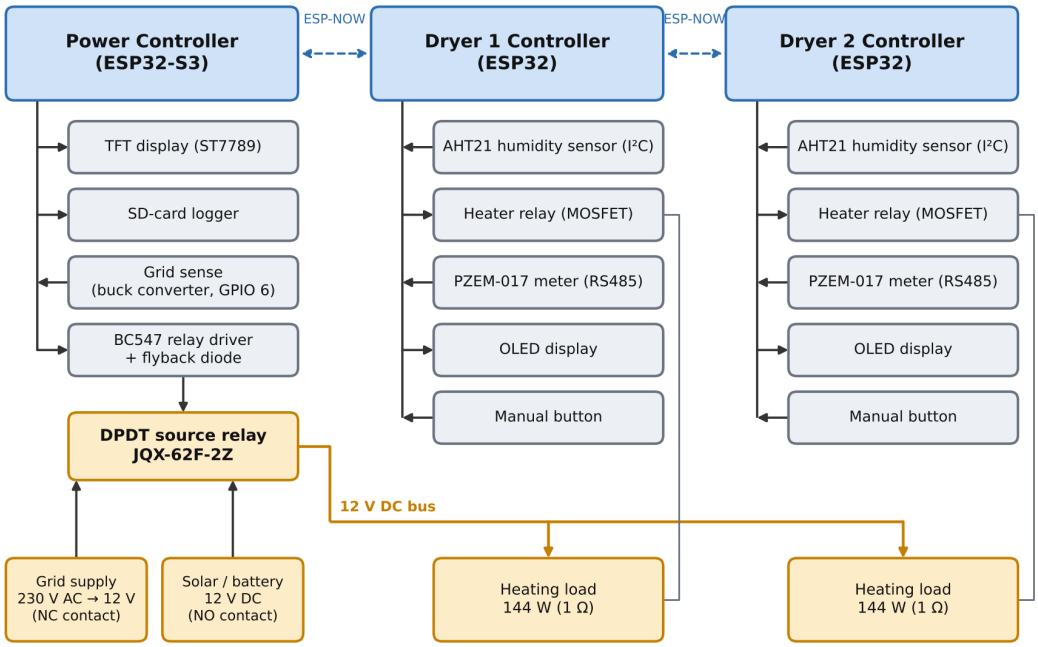
KPI	Target	Definition
Jain fairness index (JFI)	≥ 0.85	Evenness of energy (or rounds won) shared between dryers, Eq. (5)
Solar energy fraction	$\geq 65\,\%$	Share of the session supplied from solar/battery
Auction cycle time	$< 10\,\mathrm{s}$	Bid transmission to relay actuation for one round

3 System Architecture

The system has three layers (Figure 1a). **Layer 1** produces a 12 V DC bus from grid or solar/battery supply. **Layer 2** is three ESP32 boards that decide, every round, which dryer receives power. **Layer 3** is the dryers, each measuring its own humidity and power and switching its own heating load. Figure 1b shows how the three controllers connect to their sensors, relays, displays and loads.



(a) Three-layer architecture.



(b) Hardware block diagram of the three controllers.

Figure 1: System architecture.

One decision round. (1) Each dryer node reads its chamber humidity. (2) It broadcasts the reading as a bid. (3) Every node, including the power node, applies the same winner rule. (4) The winning dryer switches its heating load on and the other pauses. (5) The power node updates its display, fairness index and SD-card log. Separately, the power node monitors the grid and switches the DPDT relay whenever the grid is lost or restored.

4 Design

4.1 Layer 1 — Power infrastructure

Load analysis. Each dryer heating element is a 1 Ω, 150 W wire-wound resistor on the 12 V bus:

$$P_h = \frac{V^2}{R} = \frac{12^2}{1} = 144 \text{ W}, \quad I_h = \frac{V}{R} = 12 \text{ A}, \quad P_{tot} = 2(144) + 3(1.5) \approx 293 \text{ W}, \quad E_{day} = 293 \times 8 \approx 2.35 \text{ kWh}. \tag{1}$$

The 1.5 W terms are the three ESP32 nodes. Each dryer line has a 20 A fuse ($1.67\,I_h$), and each resistor sits on a heat sink sized for a 60 °C rise: $\theta_{HS} \le \Delta T/P = 60/144 = 0.42\,°\text{C/W}$ (a finned 100×60×40 mm extrusion gives ≈0.35 °C/W).

Full-scale solar sizing. Three 470 W panels in parallel form a 1 410 W array. With 5.5 peak sun hours (PSH) in Minna, 95 % MPPT efficiency, 3 % wiring loss and 5 % soiling loss:

$$E_{PV} = 1410 \times 5.5 \times 0.95 \times 0.97 \times 0.95 \approx 6.8 \text{ kWh/day } (\approx 2.9\,E_{day}),$$
$$I_{PV,max} = \frac{1410}{12} = 117.5 \text{ A } \Rightarrow \text{ 150 A MPPT controller.} \tag{2}$$

A 12 V, 220 Ah sealed lead-acid battery stores 2.64 kWh, of which 1.32 kWh is usable at 50 % depth of discharge (DoD). This bridges about 1320/293 ≈ 4.5 h without sun or grid. A full day of autonomy would need 390 Ah, but the array's surplus means the battery only covers short shortfalls.

Source selection. The 12 V bus is fed through a JQX-62F double-pole double-throw (DPDT) relay. With the coil off, the common (COM) contact rests on the normally-closed (NC) contact, which carries the grid supply: a 240 V AC to 12 V, 40 A switch-mode power supply (SMPS). Energising the coil moves COM to the normally-open (NO) contact, which carries solar/battery. The power node drives the coil through a BC547 transistor ($R_B = 1\,\text{k}\Omega$, so $I_B = (3.3 - 0.7)/1000 = 2.6\,\text{mA}$), and a 1N4007 flyback diode clamps the coil's switch-off spike. Because grid is the coil-off state, any controller fault leaves the dryers on grid power (*fail-safe*). Two policies were developed:

- **Solar-priority (full-scale design, simulated):** solar/battery by default, switching on battery voltage: solar at $V_{bat} \ge 12.0$ V (≈50 % SOC), grid below 11.8 V, low-voltage disconnect at 11.6 V.
- **Grid-primary (built prototype):** a buck converter steps the grid's 12 V output down to a logic level on GPIO 6. Grid present (HIGH) leaves the coil off; grid lost (LOW) energises it.

4.2 Layer 2 — Decentralised auction protocol

Nodes. The control layer has three nodes: a *power node* (ESP32-S3 with a 240×240 ST7789 TFT display and an SD-card logger) and two *dryer nodes* (ESP32 DevKit with a 128×32 OLED display and a push button). All three use ESP-NOW, Espressif's connectionless peer-to-peer radio protocol, which needs no router, network association or internet access [8].

Table 2: ESP-NOW broadcast packet and protocol timing (from firmware).

Packet fields		Timing constants	
`senderId` (uint8)	0 = power node, 1 = Dryer 1, 2 = Dryer 2	Bid interval T	10 s
`msgType` (uint8)	HELLO, BID, MANUAL_REQUEST	Stale-bid limit	25 s (older bids ignored)
`cycleId` (uint16)	Round number, so bids from one round are compared	Link timeout	30 s (node shown offline)
`rh` (float)	Chamber relative humidity (%) = the bid	Boot handshake	10 s (HELLO exchange)

Winner rule. In round k, dryer i bids its humidity $RH_i(k)$. Every node computes the same winner $w(k)$ and heater command $u_i(k)$:

$$w(k) = \arg\max_{i \in \{1,2\}} RH_i(k), \qquad u_i(k) = \begin{cases} 1, & i = w(k) \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

Because each node holds the same bids and applies the same rule, no master controller is needed and no allocation command has to be sent. Each dryer reads its own sensor, so no node can misreport its need to gain an advantage. The relay changes state only when the winner changes. Pressing a dryer's button broadcasts a MANUAL_REQUEST that triggers an immediate re-read and re-comparison ahead of the next scheduled round. The button speeds up the check but cannot override its result. Figure 2 shows the complete round.

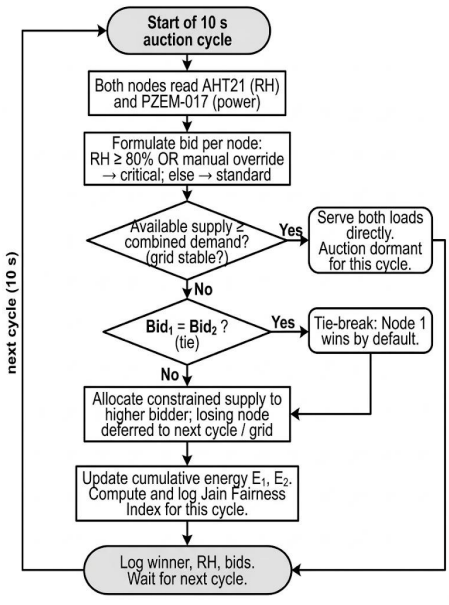


Figure 2: Auction logic for one 10 s round.

Simulation bid function. The Simulink model converts humidity into a price with a continuous, piecewise-linear bid that rises faster above the 80 % spoilage-risk threshold:

$$b(RH) = \begin{cases} 100 + \dfrac{40\,RH}{80}, & RH < 80\,\% \\ 140 + \dfrac{60\,(RH - 80)}{20}, & RH \geq 80\,\% \end{cases} \qquad \text{(units/kWh)} \tag{4}$$

Because b increases with RH, the highest bid always belongs to the most humid dryer. The hardware can therefore bid with RH_i directly and select exactly the same winner.

Fairness metric. Fairness is measured with Jain's index [9]:

$$J = \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n \sum_{i=1}^{n} x_i^2}, \qquad \frac{1}{n} \leq J \leq 1, \tag{5}$$

where x_i is the cumulative energy delivered to dryer i (simulation) or the number of rounds it has won (hardware). For two dryers, $J = 1$ is an equal share and $J = 0.5$ means one dryer has received everything. For a share ratio $r = x_1/x_2$, Eq. (5) becomes $J = (1 + r)^2/\big(2(1 + r^2)\big)$, so $J = 0.90$ corresponds to $r = 0.5$, a 1 : 2 split.

4.3 Layer 3 — Dryer endpoints

Each dryer station has an AHT21 humidity/temperature sensor (I^2C) in the chamber, a heating load switched by a 12 V relay through an N-channel MOSFET, and a PZEM-017 DC energy meter (50 A shunt) read over RS485 (Modbus RTU via a MAX485 transceiver, power register 0x0003). Energy per dryer accumulates from the metered power:

$$P_i = V_i I_i, \qquad E_i = \sum_k P_i(k)\,\Delta t. \tag{6}$$

Table 3: Hardware: full-scale design and what was built for the prototype, with node pin assignments.

Layer	Component	Full-scale design (simulated)	Built prototype / connection
1	Grid supply	240 V AC → 12 V, 40 A SMPS	As designed; relay NC contact
1	Solar / battery	3×470 W PV, 150 A MPPT, 12 V 220 Ah SLA	Regulated 12 V bench supply emulating the solar bus; relay NO contact
1	Source relay	JQX-62F DPDT; BC547 + 1 kΩ + 1N4007	As designed; coil driven from power-node GPIO 5
2	Power node	ESP32-S3, ST7789 TFT, SD card	TFT: SCLK 12, MOSI 11, DC 13, RST 15. SD: SCK 9, MISO 8, MOSI 7, CS 4. Grid sense GPIO 6
2	Dryer nodes (×2)	ESP32 DevKit, SSD1306 OLED (I^2C 0x3C), button	Button GPIO 15 (pull-up); heater relay GPIO 2
3	Humidity sensor	AHT21: 0–100 % RH (±2 %), ±0.3 °C, I^2C	I^2C bus shared with OLED; damp-cotton humidity stimulation on the bench
3	Energy meter	PZEM-017 + 50 A shunt, Modbus address 1	MAX485: RX 16, TX 17, DE/RE 25
3	Heating load	1 Ω 150 W resistor (144 W), 20 A fuse	12 V bulbs for bench validation; resistors for full-power tests

Monitoring and logging. The power-node display shows each dryer's link status, the active source, the winner, the JFI, both humidity bids, each dryer's bid delivery rate, its cycle time and its count of late rounds. The power node logs *time, event, source, winner, JFI, RH_1, RH_2, delivery %, cycle time* to its SD card. Each dryer node logs *time, event, RH, heater state, winner, power* to on-board flash.

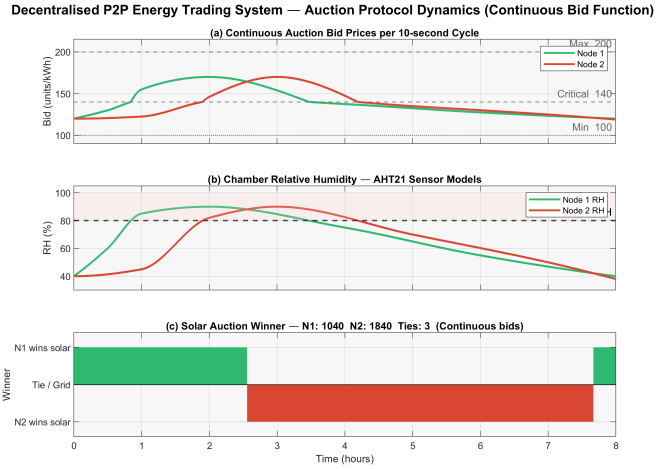
5 Development and Testing Method

Table 4: Staged development and verification.

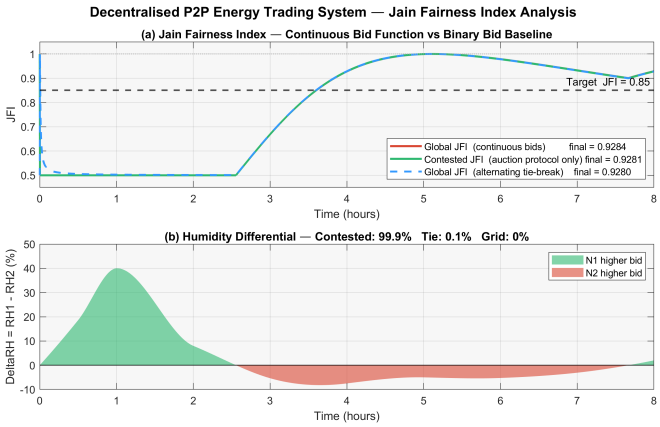
Stage	Set-up	Verified
Simulation	MATLAB/Simulink `p2p_microgrid`: PV, battery and relay subsystem; 10 s auction; two 144 W loads; ode45 solver, 8 h	Source logic, auction, fairness and solar fraction at full scale; solar-only, grid-only and weak-grid scenarios
Proof of concept	Raspberry Pi Pico with two AHT21 sensors and two relay-switched bulbs	Winner and fairness logic; found and fixed a JFI timing error and a 2 % sensor offset
Firmware	Three ESP32s on USB power with injected humidity values	ESP-NOW links, boot handshake, winner rule for all humidity pairings, JFI accuracy
Bench system	Enclosure with displays and buttons, SMPS + bench supply, 12 V bulbs	End-to-end bidding, relay switching, manual re-check, grid-loss switching, display and SD logging

6 Simulation Results (MATLAB/Simulink)

The full-scale design was simulated for 8 h (28 800 s, 2 880 rounds) with the 1 410 W array and a bell-shaped Minna irradiance profile peaking near 1 050 W/m^2. The battery started at 80 % state of charge (SOC), and each dryer followed its own humidity profile (Dryer 1 peaking ≈92 % at hour 1, Dryer 2 ≈91 % at hour 3).

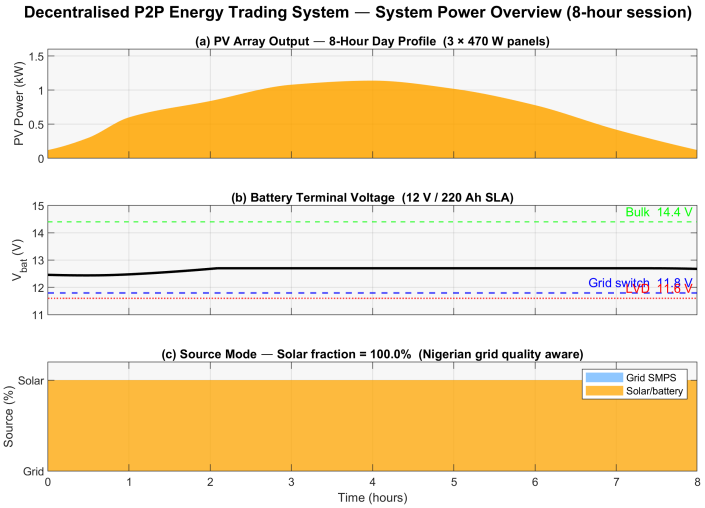


(a) Bids, chamber humidity and winner.

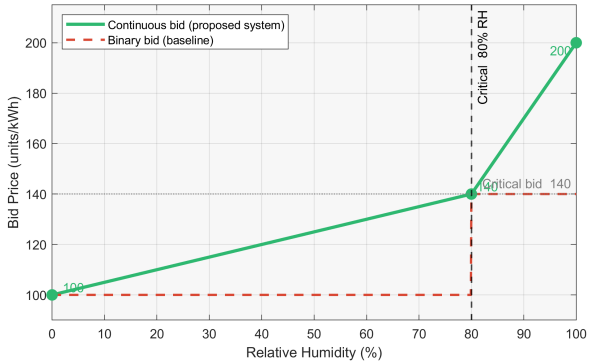


(b) JFI and humidity difference $RH_1 - RH_2$.

Figure 3: Auction and fairness over the 8 h session. Power follows the wetter dryer: Dryer 1 wins until about 2.5 h, then Dryer 2 takes over (wins: 1 040 vs. 1 840; ties: 3). The JFI stays near 0.5 while only one dryer is urgent, rises once Dryer 2 starts winning, and ends at 0.928. The tie-break rule changes the final JFI by less than 0.001.



(a) PV output, battery voltage and supply source.



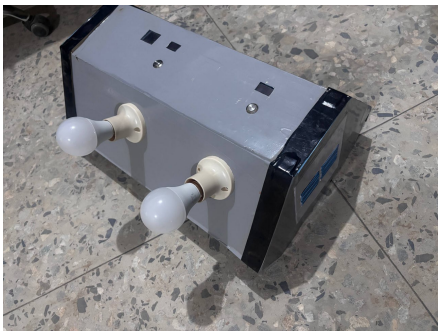
(b) Continuous bid function vs. two-level baseline.

Figure 4: Power layer and bid design. PV output peaks at about 1.05 kW, and battery voltage stays between about 12.45 and 12.7 V, above the 11.8 V grid-switch level. The dryers therefore run entirely on solar/battery (solar fraction 100 %).

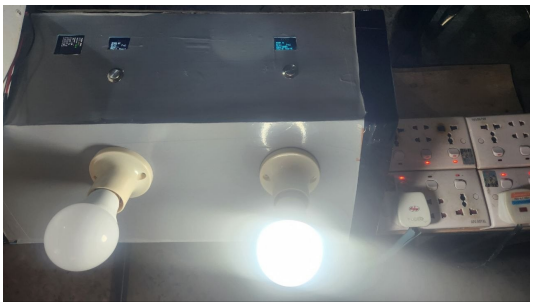
Table 5: Simulation results.

Rounds simulated / contested / ties	2 880 / 99.9 % / 0.1 %	Fairness comes from the bids themselves, not from tie-breaking
Wins Dryer 1 / Dryer 2	1 040 / 1 840	Winner switches when Dryer 2 becomes wetter
Final JFI (global / contested / alternate tie-break)	0.9284 / 0.9281 / 0.9280	Target $\geq$0.85 met
Solar energy fraction	100 %	Target $\geq$65 % met
Continuous vs. two-level bid	$\Delta J = 0.0003$	Main benefit is a smoother price signal near the 80 % threshold
Scenario: grid only, no PV	SOC 80 %$\rightarrow$0 in $\approx$1.8 h	Confirms the battery alone cannot carry the load
Scenario: weak grid with PV	Tracks solar-only case	Logic correctly prefers solar over a weak grid

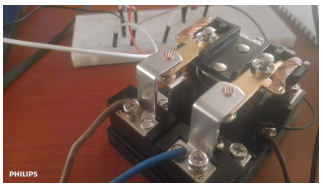
7 Prototype and Hardware Results



(a) Finished enclosure: displays, buttons, dryer outputs.



(b) Running: Dryer 2 heating, Dryer 1 paused; TFT and both OLEDs live.



(c) JQX-62F DPDT source relay.

Figure 5: Physical prototype. Bulbs stand in for the dryer heating loads during bench testing.

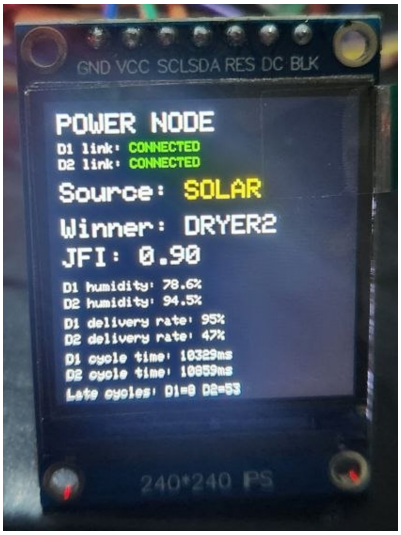


Table 6: Live readings from the power-node display: run A (photo, left) and run B.

Reading	Run A	Run B
Dryer links	both connected	both connected
Supply source / winner	solar / Dryer 2	solar / Dryer 2
Humidity D1 / D2 (%)	78.6 / 94.5	81.6 / 83.5
JFI (rounds won)	**0.90**	0.50
Bid delivery D1 / D2 (%)	95 / 47	98 / 84
Cycle time D1 / D2 (s)	10.33 / 10.86	10.35 / 10.83
Late rounds D1 / D2	0 / 53	0 / 3

Interpretation.

- **Correct, unanimous decisions.** In both runs Dryer 2 had the higher humidity, and all three boards selected it and switched accordingly. The decentralised rule works without a master controller.
- **Fairness follows need.** In run A, $J = 0.90$ means the dryers won rounds in a 1 : 2 ratio. In run B, Dryer 2 was wetter throughout, so it won every round and $J = 0.5$. This is the intended behaviour: power goes to the wetter produce, and the index rises once the dryers' humidity levels cross, as the simulation shows.
- **Source switching.** With the grid signal removed the relay moved the bus to the solar input, and it returned to grid when the signal was restored.
- **Timing and radio link.** Rounds took 10.3–10.9 s: the 10 s bid timer plus processing and transmission. Dryer 2's lower bid delivery (47 % in run A) and its late rounds point to a weaker radio path to that node, not to a logic error.

8 Performance Against Targets

Table 7: KPI results.

KPI	Target	Simulation	Hardware	Status / next action
Jain fairness index	≥ 0.85	0.928	0.90 (run A)	Met; confirm over a full logged session
Solar energy fraction	$\geq 65\,\%$	100 %	n/a (bench supply)	Met in simulation; needs a real PV array to test on hardware
Auction cycle time	$< 10\,\mathrm{s}$	fixed input	10.3–10.9 s	Close; start rounds on bid arrival and improve antenna placement

9 Innovation and Contribution

- **Need-based allocation.** Power is allocated by measured crop humidity rather than by price, arrival time or a manual timetable. Because the bid comes from a sensor, it is hard to inflate.
- **Fully offline consensus.** Three ESP32s agree on every decision over ESP-NOW, with no router, internet, cloud or blockchain, which suits rural sites [7].
- **Validated in simulation and hardware.** Most P2P energy studies are simulation-only [5, 7]. Here the same rule is shown in Simulink and on working hardware with live fairness monitoring.
- **Fail-safe hybrid supply.** The relay's coil-off state is grid supply, so a controller fault never leaves the dryers unpowered.
- **General-purpose.** Any load that can report a need signal can join as a node, so the same firmware pattern scales from two dryers to a whole community (Section 10).

10 Beyond Dryers: Community Energy Sharing and Grid Resilience

Priority sharing when supply is short. The auction never relies on the load being a dryer. It only needs each node to report a need signal, and every node to apply the same rule. For a community with mixed loads, each node's bid combines a *class weight* w_i, agreed once by the community, with a measured *need* $n_i \in [0, 1]$:

$$b_i(k) = w_i\, n_i(k), \qquad \text{serve nodes in order of } b_i \text{ until } \sum P_i \leq P_{avail}(k). \tag{7}$$

Table 8: Illustrative community priority scheme (weights set by the community, not fixed by the system).

Node	Weight w_i	Need signal n_i (examples)
Hospital / clinic	1.0	Theatre or oxygen equipment in use; vaccine-fridge temperature; emergency call
Water pumping	0.8	Storage-tank level
School	0.6	Classes in session (timetable); laboratory sessions
Homes (residential cluster)	0.4	Battery level; evening lighting demand
Mosques and churches	0.3	Scheduled prayer or service times
Agricultural dryers	0.5	Chamber humidity (as in this project)

When there is enough power, every load is served and the auction stays dormant. When supply falls, the most critical loads (the hospital first) are protected automatically and the rest share what remains. The Jain index makes sure lower-priority users are not starved over a whole day. Because the need signals come from sensors and schedules rather than requests, no user can jump the queue.

Decentralisation as an answer to grid failure. A central control room or cloud server is a single point of failure: when it, the internet or the grid goes down, every connected load goes down with it. In this design, each node holds the rule and the latest bids itself. If the national grid collapses, the community's solar-battery supply keeps running as an island, and the nodes continue to share it with no outside connection. If one node fails, the others carry on. Many independent, self-coordinating mini-grids of this kind make a community resilient to the repeated national grid collapses.

11 Applications, Limitations and Future Work

Applications. Farmer cooperatives and agro-processing centres that share dryers; rural and peri-urban communities with a shared solar mini-grid; estates, campuses and hospital compounds that must ride through grid outages; cold rooms, irrigation pumps and market clusters.

Limitations. The prototype uses a bench supply in place of the PV array and battery, so solar fraction and battery-voltage switching are validated in simulation only. Bench tests used bulbs and damp-cotton humidity

stimulation rather than full heaters and produce. The power node keeps the last winner on screen when a dryer goes offline. The BC547 is rated for 100 mA, so the relay-coil current must be checked against it.

Future work. Full-power trials with the 144 W loads and real produce; full-session fairness measurement; a real PV array and battery; an offline-winner reset; a community pilot with mixed loads (clinic, school, homes); a mobile dashboard for cooperative managers. The concept is being developed into a product, **AutaDry**.

12 Conclusion

The project shows that low-cost ESP32 boards can share scarce power fairly and automatically without internet access. In the dryer test case, the wetter dryer is served first, the fairness target is met in simulation (0.93) and on hardware (0.90), and the supply falls back safely to grid if the controller fails. The same decentralised mechanism can give a whole community priority-based, outage-resilient access to limited energy.

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